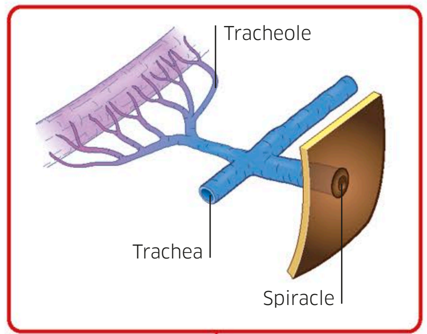
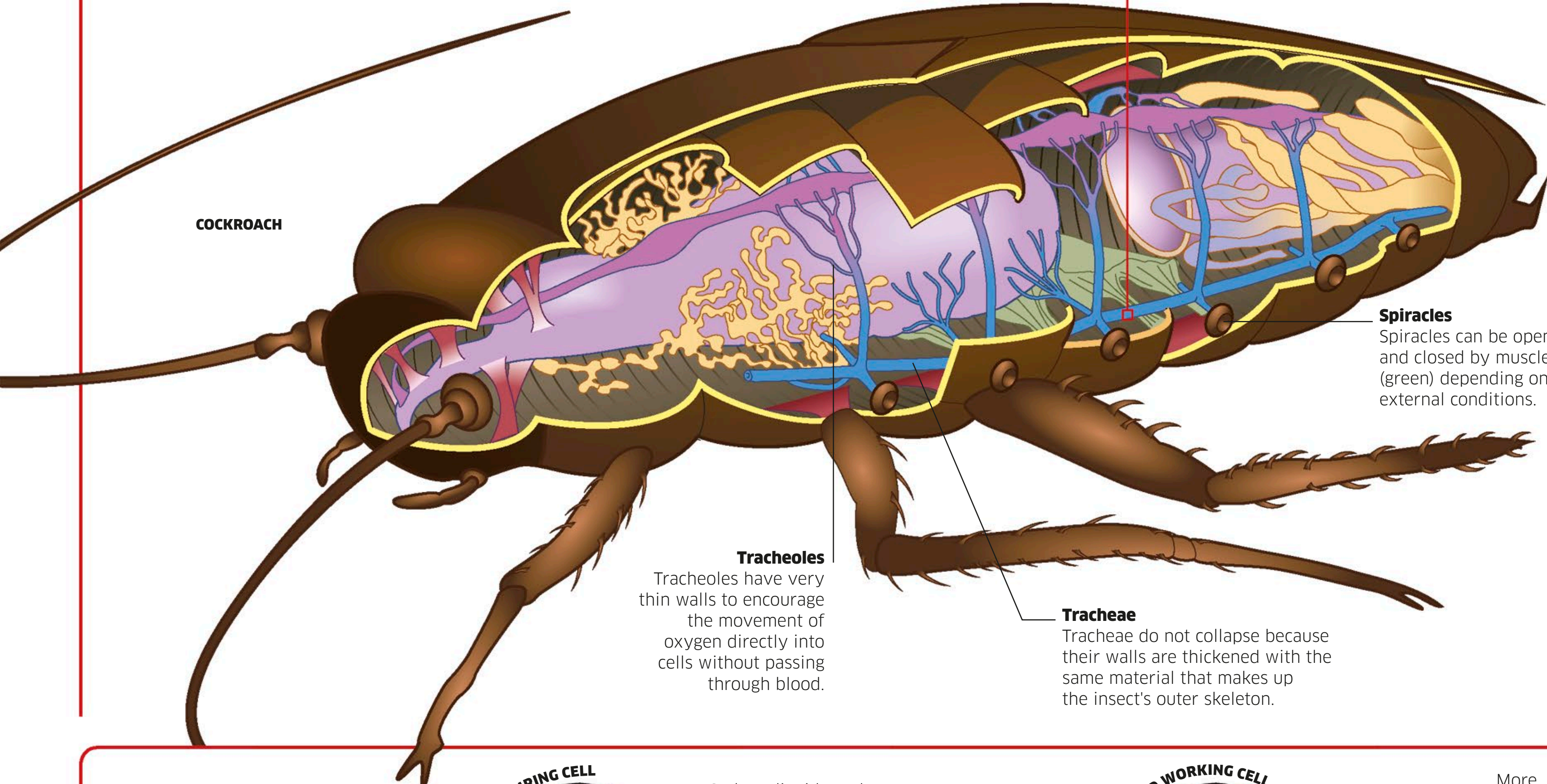
Breathing with tracheae

Insects and related invertebrates have a breathing system that gets oxygen directly to their muscles. Instead of oxygen being carried in the blood, an intricate network of pipes reaches into the body from breathing holes called spiracles. Each pipe—known as a trachea—splits into tinier branches called tracheoles. The tracheoles are precisely arranged so that their tips penetrate the body's cells. This delivers oxygen-rich air from the surroundings deep into the insect, where respiration takes place.



Tracheoles
Microscopic air-filled tracheoles in the body of an insect perform a similar role to blood-filled capillaries in other animals: they pass oxygen into the cells, while carbon dioxide moves out. This direct and efficient system means cells get oxygen delivered straight to them.

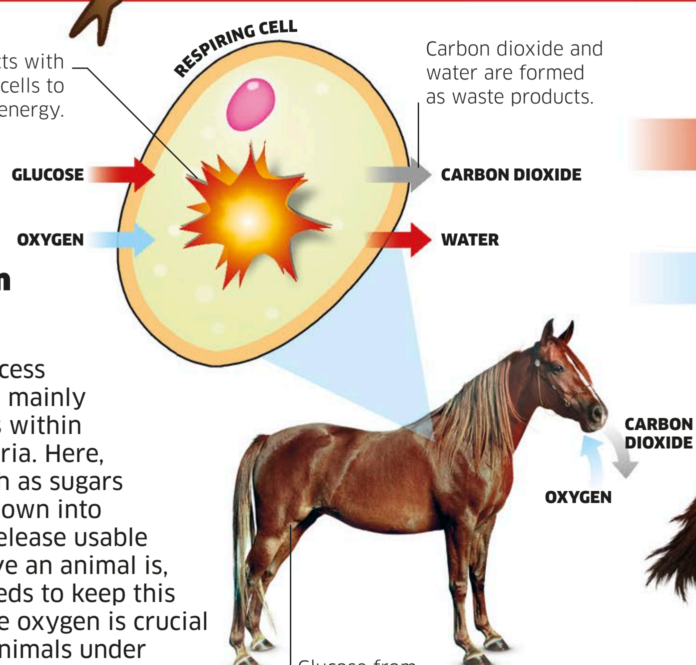


Spiracles
Spiracles can be opened and closed by muscles (green) depending on external conditions.

Tracheoles
Tracheoles have very thin walls to encourage the movement of oxygen directly into cells without passing through blood.

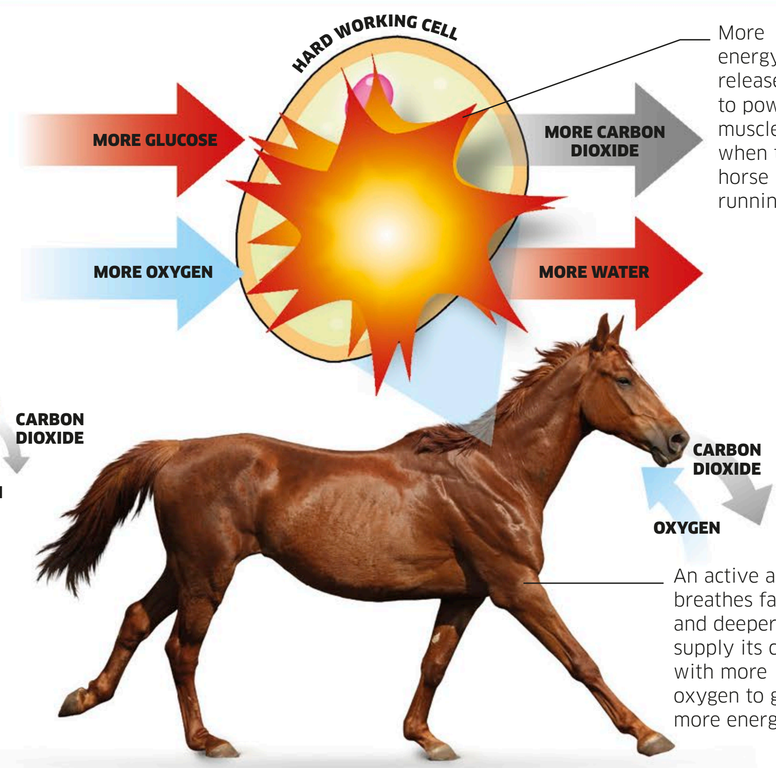
Tracheae
Tracheae do not collapse because their walls are thickened with the same material that makes up the insect's outer skeleton.

Glucose reacts with oxygen inside cells to release energy.



Carbon dioxide and water are formed as waste products.

Glucose from digested food is stored inside cells until it is needed.



More energy is released to power muscles when the horse is running.

An active animal breathes faster and deeper to supply its cells with more oxygen to get more energy.

Cellular respiration

The oxygen an animal brings into its body is used in a chemical process called respiration. This mainly takes place in capsules within cells, called mitochondria. Here, energy-rich foods, such as sugars (glucose), are broken down into smaller molecules to release usable energy. The more active an animal is, the more oxygen it needs to keep this reaction running. While oxygen is crucial for most respiration, animals under physical pressure can release a tiny bit of extra energy without oxygen—a process known as anaerobic respiration.